Ridge Broker Ablation Experiment

Question and design. These ablations probe three components of the broker's learning setup under Ridge learning. They do not provide an additive decomposition of the broker's advantage. The paired-Ridge reference pools the broker's mediated-match history and uses additive and bilinear pair features. *Size-matched pooled history* retains these features but uses the smaller of the broker's available sample and the median agent fitting-sample size. *Single-principal history* randomly retains one endpoint type from each mediated match for fitting; the endpoint labels have no directional meaning. At prediction, the fitted one-type function is applied to both parties and the fitting-sample output mean is subtracted once. This variant produces an additive pair score and cannot represent a bilinear interaction. *Additive Ridge* retains both endpoint types in every fitting row but replaces the pair features with their type-vector sum, so it also cannot represent a bilinear interaction. The agent learning rule and parameters are unchanged, but realized agent histories can change as matching evolves. All conditions use $\lambda_a = 0.003$, $\lambda_b = 0.001$, the same period window and fitting cadence, and the same broker observation cap before any additional size matching.

The primary estimand in realization c is the late-period broker-agent holdout rank-correlation gap

$$G_v(c) = \overline{r}_{b,v}(c) - \overline{r}_{a,v}(c),$$

for Ridge broker variant v. The ablation contrast is $G_v(c) - G_{\text{pair}}(c)$. Each bar averages periods 481–500 within seed and then averages seeds. The baseline comparisons use 50 common seeds. Other conditions use 20 common seeds. The $\rho \times \delta$ analysis gives one observation to each of the 26 effective realizations, including the baseline. Repeated grid coordinates that map to the same realization receive no additional weight. Because the ablations alter broker rankings, matches, and later learning histories, their contrasts are system-level effects of the specified restriction, not fixed-data comparisons of fitted regressions.

Baseline. Table RA1 reports late-period baseline outcomes. Figure RA1A shows the full baseline trajectory rather than only its late-period summary.

Table RA1: Baseline outcomes by Ridge broker variant

Outcome	Pair	Size-matched	Single-principal	Additive
Agent holdout rank correlation	0.479	0.706	0.704	0.697
Broker holdout rank correlation	0.814	0.727	0.213	0.231
Broker-agent rank-correlation gap	0.335	0.021	-0.492	-0.466
Brokered-minus-self output gap	0.530	0.390	0.182	0.175
Outsourcing rate	0.939	0.696	0.339	0.364

Entries use all 50 common seeds. Each seed is first averaged over periods 481–500.

Results across effective realizations. Table RA2 summarizes the primary outcome over the baseline and $\rho \times \delta$ design. Figure RA1B–D shows the paired realization contrasts, while Figure RA2 displays the $\rho \times \delta$ pattern for each broker variant.

Table RA2: Broker-agent rank-correlation gap across effective realizations

Variant	Median gap	Median Δ	Δ [P10, P90]	Gap > 0	$\Delta > 0$
Pair	0.458	–	–	26/26	–
Size-matched	0.065	-0.391	[-0.476, -0.205]	20/26	0/26
Single-principal	-0.344	-0.629	[-1.231, -0.294]	6/26	0/26
Additive	-0.329	-0.608	[-1.215, -0.112]	7/26	1/26

Δ is the ablation-minus-pair contrast. Each effective realization receives one observation.

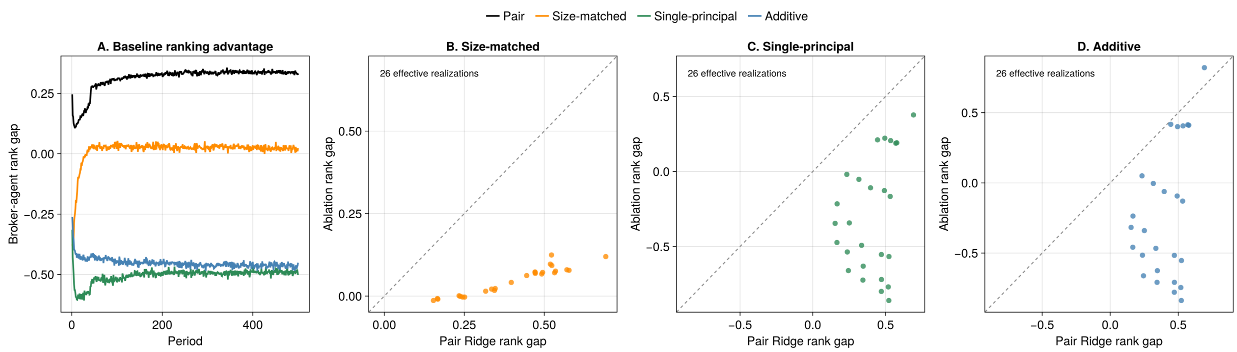

Figure RA1: Ridge broker ablations. Panel A uses all 50 baseline seeds. Panels B–D give one point to each effective realization. Dashed lines mark equality with paired Ridge.

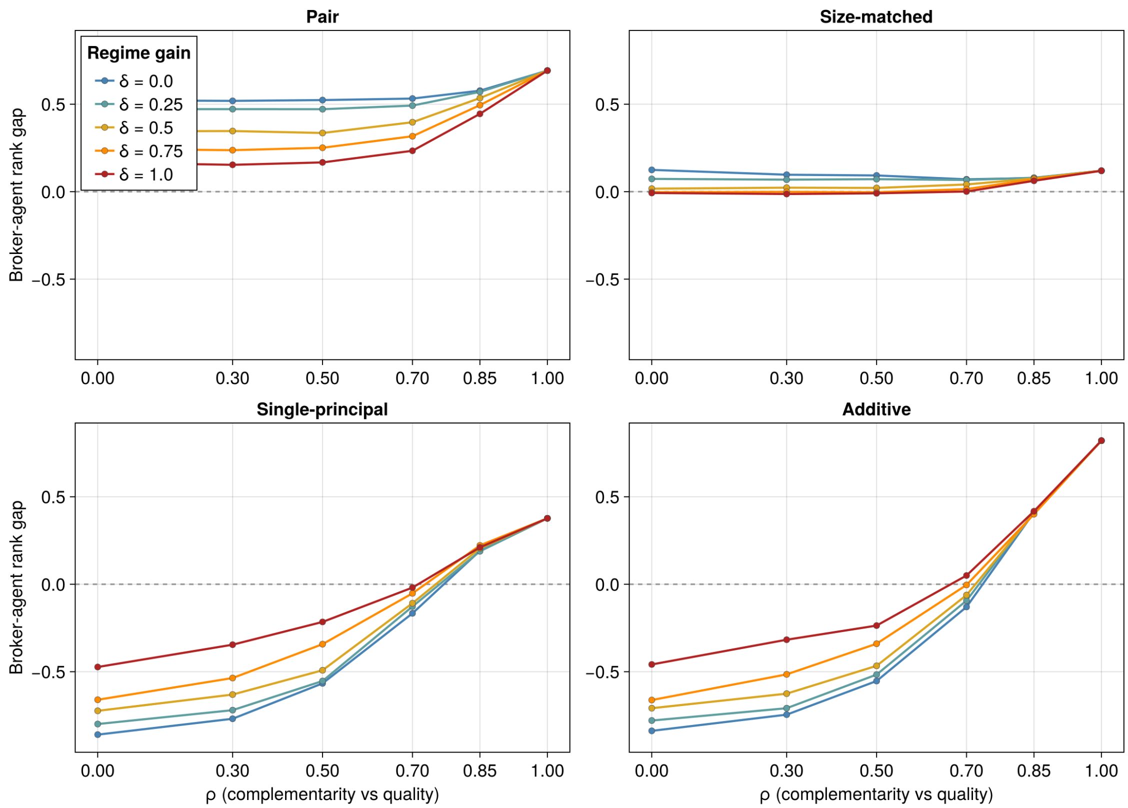

Figure RA2: Broker-agent rank-correlation gaps across the $\rho \times \delta$ design. Each panel shows one Ridge broker variant, each line shows one value of δ, and markers are late-window seed means. The vertical scale is shared across panels, and the dashed line marks a zero broker-agent gap. At $\rho = 1$, δ is inactive, so the coincident endpoints represent one effective realization and receive no additional analytical weight.

Interpretation. At the baseline, every ablation reduces the broker's ranking advantage. The size-matched gap is 0.021, a paired change of -0.314 from paired Ridge (Monte Carlo standard error 0.007). The single-principal and additive gaps are -0.492 and -0.466, with paired changes of -0.827 and -0.802. Their Monte Carlo standard errors are 0.023 and 0.021.

The size-matched contrast is consistent across the design. Its rank gap is lower than paired Ridge in all 26 effective realizations. The median change is -0.391, and its 10th and 90th percentiles are -0.476 and -0.205. The gap nevertheless remains positive in 20/26 realizations. The gap change is not solely a change in the broker's prediction. The median broker rank correlation changes by -0.082, while the median agent rank correlation changes by +0.303. The agent rank correlation also exceeds paired Ridge in all 26 realizations under size matching, all 26 under single-principal history, and 24 under additive Ridge. This pattern is consistent with an exploration mechanism: altered broker rankings change the partners agents encounter and may broaden the diversity of partner types in their learning histories. The saved diagnostics do not measure that diversity, so these results do not establish the proposed mechanism directly.

The pair-versus-additive contrast holds both endpoint types in each broker fitting row while removing the bilinear pair features. As Figure RA2 shows, the system-level effect is larger when the interaction component of the production function is strong. The median additive-Ridge gap rises from -0.709 at $\rho = 0$ to 0.411 at $\rho = 0.85$ and 0.821 at $\rho = 1$. At $\rho = 1$, where δ drops out and the additive representation is correctly aligned with the production function, additive Ridge exceeds the paired-Ridge gap of 0.692. The latter result is consistent with a finite-sample cost from unnecessary bilinear regressors, but the rank-gap contrast alone does not establish that mechanism because matching and agent learning also adjust.

Single-principal and additive Ridge both produce additive pair scores, but they use different fitting rows. Additive Ridge retains $x_i + x_j$ in each row; single-principal Ridge retains one randomly selected endpoint and reconstructs a pair score by adding two fitted one-type scores. Their baseline paired rank-gap difference (single-principal minus additive) is -0.025, with Monte Carlo standard error 0.018. Across effective realizations, the difference has median -0.027 and 10th and 90th percentiles -0.213 and -0.004; it is positive in only 2/26 realizations. Thus, among the two additive-score variants, retaining both endpoint types in each fitting row produces a modest system-level ranking benefit in most realizations. This is not a comparison of bilinear representations: neither variant can represent the bilinear interaction, which is strongest at low ρ.

The observed brokered-minus-self output gap remains positive in all 26 realizations under each ablation, even when the broker's rank gap is negative. This channel mean difference is descriptive, not a causal estimate of brokerage's effect on output. Agents endogenously choose channels, the channels expose them to different candidate sets, and acceptance decisions differ. The positive gap therefore does not show that brokerage improves a match while holding the demander and feasible candidates fixed, nor does it identify access, selection, or network position as its cause. The ablation contrasts are full-system interventions: prediction, matching, learning histories, outsourcing, and network position adjust together. They estimate the consequences of each modeled restriction for the resulting system, not additive shares of a fixed-data prediction advantage.